Option 1: Smart Grocery Store Billing & Insights System

Context:
A local grocery store wants to automate billing and understand sales trends.

Objective:
Build a Python program to handle multiple customer purchases, calculate bills with discounts, and generate insights.

Key Requirements:

- Maintain product catalog (name, category, price, stock)
- Process multiple customer orders with multiple items
- Validate product availability and stock
- Apply pricing rules:
    - 10% discount if total bill > ₹1000
    - 5% discount if quantity of a product > 10
    - Add 5% GST
- Update stock after each purchase
- Generate insights:
    - Total revenue
    - Most sold product
    - Category-wise sales
    - Remaining stock

Option 2: Food Delivery Order Management System

Context:
You are building the core backend logic of a food delivery app like Swiggy/Zomato.

Objective:
Create a system to manage menus, process orders, apply offers, and track customer activity.

Key Requirements:

- Store restaurant menu (item name, category, price)
- Allow multiple users to place orders with multiple items
- Calculate bill with:
    - Delivery charges based on order value
    - Coupon discounts (e.g., FLAT100, 10% OFF)
- Handle order history for each user
- Generate insights:
    - Total orders placed
    - Most ordered item
    - Highest spending customer
- Validate invalid inputs (wrong item, invalid coupon)

Option 3: Movie Ticket Booking System

Context:
A theatre wants to digitize ticket booking similar to BookMyShow.

Objective:
Develop a Python system to manage movies, seat booking, and pricing.

Key Requirements:

- Maintain movie list with timings and seat availability
- Allow users to:
    - Select movie and showtime
    - Book multiple seats
- Implement seat validation (already booked / invalid seats)
- Pricing rules:
    - Higher price for weekends or prime time
    - Discount for booking more than 5 tickets
- Support booking cancellation and seat release
- Generate insights:
    - Total bookings
    - Revenue per movie
    - Most popular show

General Instructions:

- Use functions for modular design
- Use dictionaries and lists for data handling
- Apply loops and conditions effectively
- Write clean, reusable, and readable code